Q-SOC: A Quantum-Aware Security Operations Center Prototype for Threat Detection and Post-Quantum Readiness Assessment

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***Abstract*—Security operations centers (SOCs) are correlating events from mixed sources; while organizations are commonly being asked to inventory their cryptography ahead of post-quantum migration. In this broader context, q-SOC is, in essence, any prototype that places both functions behind one API and one dashboard – the rule-based detection engine with one controlled attack simulator, one more cryptographic-asset assessment with a composite Quantum Readiness Index (QRI), educational Qiskit demonstrations of Grover's and Shor's algorithms, and ML-DSA-65 report signing; the framework is being outlined by this article architecture and data pipelines as realized, reports the outputs of a demonstration run of the application as tables and graphs, and gauges the prototype with trials run against the unmodified backend. On 11 simulator scenarios, 220/220 is running raised the anticipated alert, and a negative control with the rule disabled failed 33/33. Median ingestion throughput over HTTP on a single-process SQLite deployment rose from 35 events/s (single-event requests) to 1526 events/s (500-event batches). A robustness check found that 6 of 37 malformed inputs caused server errors, and an examination of the QRI disclosed that its ranking is principally invariant to the weights and that unclassified algorithms are scored inconsistently across the parts. Within this framework, all data are, in this setting, synthetic and from a person machine, and we do not claim production readiness, real-world detection accuracy, or certification.**

***Index Terms*—security operations center, SIEM, post-quantum cryptography, cryptographic inventory, quantum readiness, detection validation**

I. Introduction

The security operations center (SOC) is the organizational unit that monitors, detects, triages and responds to security events. It is combining people, processes and technology [1]. Its central technical constituent is generally any security information and event management (SIEM) platform, which is collecting and normalizes logs from many sources, connects them against detection logic and introduces alerts to analysts; the capabilities, trends and employ of SIEM, spanning in critical infrastructures, have been reviewed in [2]. Detection logic ranges from rule engines, whose lineage is encompassing event-driven network monitors such as Bro [3], to the statistical and machine-learning techniques surveyed in [4], [5]. In this broader context, alerts are, in essence, then enriched with threat intelligence [6] and labelled with shared vocabularies such as MITRE ATT&CK, which has also been applied to structure organizational risk assessment [7].

In parallel, progress in quantum computing is changing the long-term threat model for the cryptography that a SOC is meant to protect. Shor's algorithm [8] is revealing that a Integers and com- can be factored by large enough quantum computer pute discrete logarithms in polynomial time, thereby breaking RSA and elliptic-curve public-key approaches, whereas Grover's One more quadratic speed-up against symmetric is being only supplied by search keys and hash preimages. Resource estimates indicate that a 2048-bit RSA modulus could be factored in about eight hours with roughly twenty million noisy physical qubits [9]. Present devices are remaining in the noisy intermediate-scale regime [10], so the exposure is, in essence, future and not current. It is still operationally applicable, including encrypted data that must stay confidential for many years can be collected now and decrypted later, and the direct time needed to replace cryptography across an organization is long.

The response is uniformly post-quantum cryptography (PQC), which replaces vulnerable primitives with methods believed to resist quantum attack [11]. Lattice-based strategies in particular have been surveyed from one implementation perspective [12], and ML-KEM, ML-DSA and SLH- have been standardized by NIST DSA (FIPS 203, 204 and 205). Migration needs knowing where cryptography is, in practice, employed, prioritizing assets and sequencing change, as analyzed for organizations [13], for software systems [14] and for blockchains [15]. Quantum key distribution is a separate, physics-based line of work [16], [17]; Q-SOC does not apply it.

Granted that both threads concern the same assets, the tooling is typically separate. SOC dashboards present events, alerts and incidents; while cryptographic-readiness assessments live in spreadsheets or one-off inventories, so analysts have no single view of present-day attacks and future quantum exposure. Work that brings quantum computing into the SIEM has so far used quantum algorithms to accelerate analytics [18]. A diverse question is asked, namely can a cryptographic-readiness assessment are sharing one operational surface with standard, rule-based detection, and what does such a prototype

look like when its behaviour is measured rather than outlined?

Q-SOC is regularly our attempt at an answer, and is a research prototype with two layers behind one FastAPI backend and one React dashboard. The SIEM layer ingests events in five formats, normalizes them, runs 17 transparent detection rules, raises alerts tagged with MITRE ATT&CK techniques and supports incident handling with an append-only timeline. A controlled simulator is generating synthetic telemetry for 11 cases and verifies that the anticipated alert for each person situation is triggered as anticipated. An inventory of cryp- is kept by the quantum layer tographic assets, classifies each algorithm against Shor- and Grover-style exposure, obtains a readiness status, combines the findings into the QRI, and is running small Qiskit demonstrations of Grover's and Shor's algorithms on a local simulator. We Questions are threed by study. *RQ1:* does the detection pipeline behave as stipulated, and is the validation check meaningful? *RQ2:* what are its throughput and latency, particularly for failure modes on modest hardware? *RQ3:* how does the composite readiness index behave, and where is it misleading?

The insights are, in essence, deliberately modest. (i) We document the architecture and the two data pipelines from the code (Section III). (ii) We report the outputs of a demonstration run of the application as tables and graphs (Section V). (iii) We outline a reproducible evaluation harness that imports the real backend instead of re-implementing it, and report measured findings including the shortcomings they exposed (Sections VI and VII). (iv) We separate what was run from what was not (Section VIII). We do not claim novelty over the literature, we do not claim that Q-SOC detects real attacks better than any existing product, and we make no claim about breaking real cryptography. The remainder of the paper reviews related The order of the advancements is then followed by work (Section II), above and ends with discussion and conclusions.

II. Related Work

Table I positions the 19 journal articles we depend on. For SOC and SIEM, the systematic study in [1] and the SIEM review in [2] describe the functions a SOC platform is anticipated to offer, which Q-SOC reproduces at prototype scale. On detection, the machine-learning surveys [4], [5] and the dataset survey [19] explain why we follow the transparent-rule approach substantiated against labelled synthetic situations, rather than a learned model whose precision would hinge on data we do not have; [3] is any early example of event-driven, policy-based network intrusion detection. Threat-intelligence employ [6] motivates the indicator-match rule, and ATT&CK usage [7] motivates tagging every alert with one scheme.

On the quantum side, [8] and [9] define the exposure that Q-SOC classifies, and [10] places today's hardware, which is explaining why our Qiskit demonstrations are educational. In parallel with this, PQC overviews and implementation surveys [11], [12] and migration studies [13], [15] motivate the asset-inventory and readiness-status design. The quantum-key-distribution reviews [16], [17] are included for completeness and to delimit scope. Closest to our setting, Tehrani et al. propose a second quantum-enhanced SIEM architecture for botnet detection [18]; their Quantum computation is applied by work to accelerate analytics, whereas Q-SOC is keeping detection standard and tackles the cryptographic-readiness side of the same operations center. We found no work that gauges a SOC prototype that also scores cryptographic readiness, but our search was not systematic and Rule such work out is cannoted. Shor's and Grover's algorithms, NIST FIPS 203/204/205, Qiskit and MITRE ATT&CK are tools or standards the prototype uses; where a journal source exists it is specifically cited, and otherwise they are named.

A. Technical Background

The quantum layer of Q- is supported by three technical notions SOC; we state them here so that later sections can, in practice, refer to them rigorously.

Shor-type exposure. Shor's algorithm [8] is finding the period r of $f(x) = a^x \bmod N$ in time polynomial in $\log N$, at least in practice. When r is broadly even and $a^{r/2} \not\equiv -1 \pmod N$, a non-trivial factor is following from $\gcd(a^{r/2} \pm 1, N)$. For the toy instance employed in Q-SOC ($N = 15$, $a = 7$) the order is generally $r = 4$, so $a^{r/2} = 49$ and $\gcd(48, 15) = 3$, $\gcd(50, 15) = 5$. The same period-finding structure yields discrete logarithms, which is particularly why RSA, finite-field Diffie, Hellman and elliptic-curve schemes are routinely grouped as one exposure class.

Grover-type exposure. A marked item is found by grover's search among M candidates with about $\frac{\pi}{4}\sqrt{M}$ oracle calls. Applied to a k-bit symmetric key this halves the effective exponent, from 2^k to roughly $2^{k/2}$ operations, and the same val weakens hash preimage resistance. The effect is a reduced margin and not a break; so Q-SOC labels AES-128 and SHA-256 as "review" and AES-256 and SHA-384 as lower exposure, and does not claim that Grover's algorithm "breaks AES".

Readiness is principally not risk. A prop- are summarized by the risk class erty of an algorithm family; whereas readiness is outlining an organization's state for one asset. Q-SOC therefore is keeping two axes, namely a risk classification (HIGH, REVIEW, LOWER, UNKNOWN) and a readiness status (NOT_ASSESSED, LEGACY, AT_RISK, PARTIALLY_READY, PQC_READY, UNKNOWN). Table II obtains. The status PQC_READY is never obtained from a risk class; it is frequently claimed only when the recorded algorithm name is itself a acknowledged post-quantum primitive (ML-KEM, ML-DSA, SLH-DSA or their pre-standard names).

The index. In light of this dynamic, with n assets of which n_H are systematically HIGH, the crypto-readiness component behaves, on catalog-only inventories, as $\mathrm{CR} = 100\,(1 - n_H/n)$. Quantum exposure QE and migration preparedness MP are specified in the code from the same category, and

$$\mathrm{QRI} = w_c\,\mathrm{CR} + w_m\,\mathrm{MP} + w_e\,\mathrm{QE} \tag{1}$$

with $(w_c, w_m, w_e) = (0.4, 0.3, 0.3)$. The weights are predominantly experimental defaults that the interface labels as a

TABLE I
POSITIONING OF THE CITED WORK. ALL ENTRIES ARE JOURNAL ARTICLES; VENUE QUARTILES ARE SCIMAGO SJR 2025 (SEE SECTION VIII).

Ref.	Venue (year)	Topic	Relevance to Q-SOC
[1]	IEEE Access (2020)	SOC systematic study, open challenges	Frames the people–process–technology scope of a SOC
[2]	Sensors (2021)	SIEM analysis, trends, critical infrastructures	Baseline SIEM functions reproduced at prototype scale
[7]	Sensors (2021)	MITRE ATT&CK in risk assessment	Motivates ATT&CK tags on alerts
[19]	Comput. & Security (2019)	Network IDS datasets survey	Data limits; why labelled synthetic scenarios are used
[4]	IEEE Commun. Surv. Tut. (2016)	Data mining / ML for intrusion detection	Contrast: Q-SOC uses transparent rules
[5]	Cybersecurity (2019)	IDS techniques, datasets, challenges	Contrast and dataset caveats
[3]	Computer Networks (1999)	Bro: real-time network intruder detection	Precedent for event-driven rule engines
[6]	Comput. & Security (2018)	Technical threat intelligence survey	Motivates indicator-match rule
[18]	Scientific Reports (2024)	Quantum-enhanced SIEM, botnet detection	Closest work; quantum for analytics, not readiness
[8]	SIAM J. Comput. (1997)	Quantum factoring, discrete logarithms	Basis of the HIGH-risk classification
[9]	Quantum (2021)	RSA-2048 resource estimate	Scale of the exposure; toy demo is not an attack
[10]	Quantum (2018)	NISQ era	Why demonstrations are educational
[11]	Nature (2017)	Post-quantum cryptography overview	Replacement candidates in the inventory
[12]	ACM Comput. Surv. (2019)	Lattice-based PQC implementations	Context for ML-KEM / ML-DSA entries
[13]	Nature (2022)	Organizational transition to PQC	Motivates inventory and prioritization
[14]	IEEE Access (2024)	Migrating software to PQC, literature review	Motivates readiness status and roadmap
[15]	IEEE Commun. Surv. Tut. (2026)	Quantum-era risk assessment, migration (blockchain)	Risk assessment and migration framing
[16]	Rev. Mod. Phys. (2002)	Quantum cryptography (QKD) review	Scope delimiter: not used
[17]	Adv. Opt. Photon. (2020)	Advances in quantum cryptography	Scope delimiter: not used

TABLE II
DERIVATION OF READINESS STATUS FROM RISK CLASSIFICATION.

Risk class	Readiness	Rationale
HIGH	LEGACY	Public-key scheme exposed to Shor; needs a migration plan
REVIEW	AT_RISK	Reduced margin under Grover (AES-128, SHA-256)
LOWER	PARTIALLY_READY	Adequate theoretical margin, but not a PQC-native primitive
UNKNOWN	UNKNOWN	Algorithm absent from the catalog
(PQC name)	PQC_READY	Recognized ML-KEM / ML-DSA / SLH-DSA entry

TABLE III
MEASURED REPOSITORY INVENTORY (COMMIT D7AD437).

Item	Count
Detection rules (auth 4, endpoint 6, net. 3, web 3, TI 1)	17
MITRE mapping: direct / indicative / none	4 / 7 / 6
Simulator scenarios	11
Ingestion formats	5
Catalog entries (HIGH 7, REVIEW 2, LOWER 12)	21
Non-blank LOC: backend app / tests	7274 / 1718
Non-blank LOC: agent / frontend	605 / 3633
Tests passing: backend / root	193 / 17

suggested research metric; Section VII is studying how the index behaves.

III. SYSTEM ARCHITECTURE AND DATA PIPELINE

A. Architecture

Q-SOC is a three-tier application (Fig. 1). A React single-page application and event sources (an endpoint agent, log shippers and the built-in simulator) reach a FastAPI backend through an nginx reverse proxy that terminates TLS and forwards both REST calls and the live-event WebSocket. The backend is layered: thin route modules invoke services that use the ingestion normalizer, the detection engine, the modelling service and the quantum module; security dependencies (JWT, Argon2 password hashing, role-based access control and login rate limiting) secure all routes except health and login. State is broadly kept through SQLAlchemy in PostgreSQL under Docker Compose, or in SQLite for development. Table III lays out counts measured from the checkout rather than taken from documentation.

B. Detection data pipeline

Fig. 2 is, for all practical purposes, following any event from its origin to the analyst. (1) The source is sending a raw payload to `/api/events` or `/api/events/batch` in one of five formats (generic, Linux auth, Windows Security, network, simulator); only the analyst and admin roles may, in effect, write. (2) The normalizer translates the payload into one typical Event schema (timestamp, host, event type, severity, user, tackles, message, raw event). (3) The event is stored and its host record is modified. (4) The Every rule module is performed by detection engine, and each person module retrieves its own time window from the database, so detections take previous events into the account together with the current batch. (5) A further firing rule that is, broadly speaking, not a recent copy creates an Alert including rule, severity, confidence, evidence event identifiers and a MITRE ATT&CK tag. Building on this observation, alerts are, in essence, acknowledged by (6) Analysts and create Incidents that link alerts as indication; each note and status update is recorded as an append-only timeline entry. (7) The dashboard, live stream and CSV/PDF reports are reading from the same

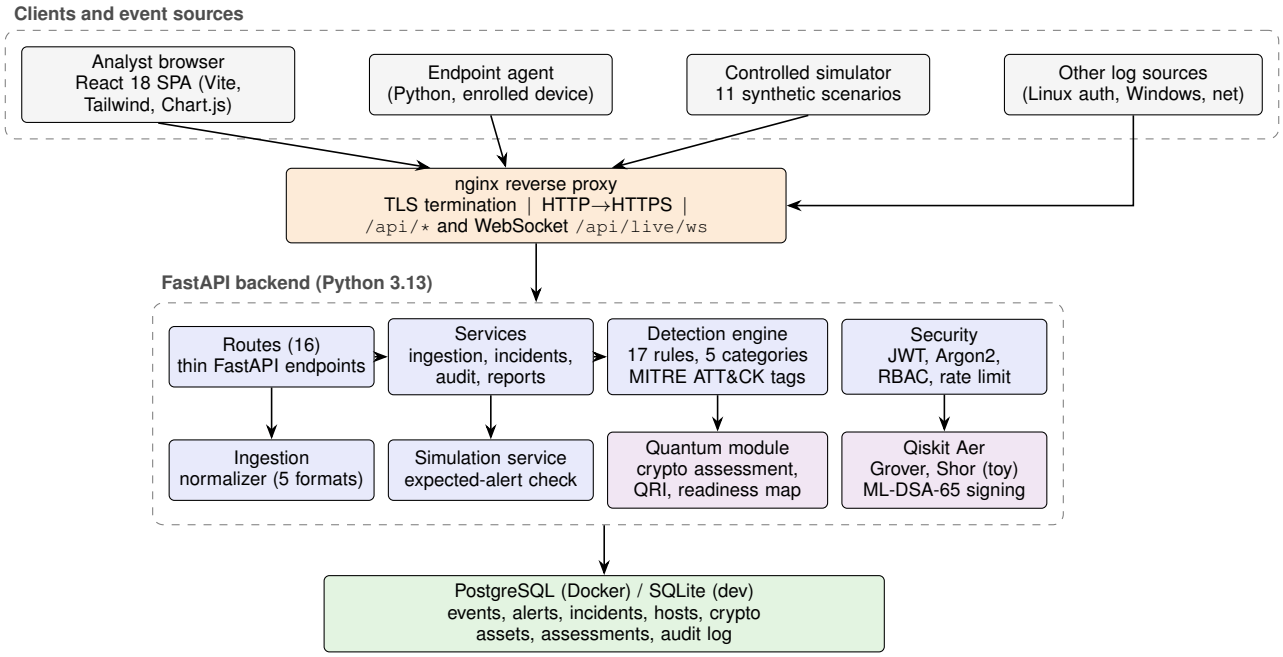


Fig. 1. Q-SOC system architecture as implemented. Blue: SIEM/SOC components; violet: quantum components; green: persistence.

tables. Detection is running synchronously within the ingest request.

C. Verification of detection and the simulator

Synthetic events with labels are created by the simulator (documentation-range tackles, 203.0.113.0/24) for 11 scenarios and is logging the related rule, severity, and MITRE tag that each should generate. A run is *passing* solely if a new alert references that run's events and all three characteristics align; It is counted as a failure, if not; indeed, a mode called "authorised lab" does not inject anything and assesses actual telemetry from a permitted, registered endpoint. Out of 17 rules, just 4 have a direct ATT&CK mapping; the code marks 7 as indicative and 6 as unmapped, meaning ATT&CK labels shouldn't be construed as claims of coverage.

D. Quantum-readiness pipeline

Cryptographic assets (algorithm, framework, host, protocol, environment, owner) are, in this context, grouped as stated by a 21-entry catalog into HIGH (public-key strategies compromised by Shor's algorithm), REVIEW, and LOWER (symmetric and hash primitives influenced by Grover's algorithm at a reduced margin). Any autonomous mapping is creating a readiness status (LEGACY, AT_RISK, PARTIALLY_READY, PQC_READY, UNKNOWN) grounded in the category and migration status. In light of this dynamic, crypto readiness (CR) is integrated by the QRI, migration preparedness (MP), and quantum exposure (QE) with experimental default weights of 0.4/0.3/0.3, signified in the interface as the proposed research metric. Given that CR and QE align in catalog-only inventories, $\mathrm{QRI} = 0.7\,\mathrm{CR} + 0.3\,\mathrm{MP}$ in that setting (verified accurately, Section VII). The Qiskit A 2-qubit space and factor N are studied by examples $= 15$ applying one more local Aer simulator; educational purposes are, in effect, served by them and are frequently not intended as attacks. The `cryptography` library is applied to sign reports with ML-DSA-65 (FIPS 204); hybrid TLS in nginx, post-quantum server certificates, and ML-DSA-signed JWTs are, in essence, marked as blocked or not realized in the repository's own status.

E. Data model

Storage is, in practice, handled via Alembic (three revisions, beginning with the quantum-only foundation and encompassing the SIEM/SOC and RBAC layer). Viewed through this lens, Table IV is summarizing the 14 tables in the database reference; 15 ORM models are, put simply, stated by the code and also encompasses alert notes, indicators of compromise, and enrolled endpoints. Two design options are, in essence, prominent. First alerts save the identifiers of their evidence events, allowing each individual alert to be linked back to the original payload that triggered it. Second, one plaintext password is spanned by no column, secret, or token: an Argon2 hash is only kept by users, and secrets are reliably stored solely in environment variables. The development During the start-up, tables are yielded by route; while the PostgreSQL route rests on `alembic upgrade head`.

F. Security model

The ingestion API is, in this setting, the one place where untrusted data enters, and the simulator is the one element that could be mistaken for offensive tooling, so both are curbed. Table V gives the role model; Table VI is, as it stands, listing the controls we found in the code and documentation.

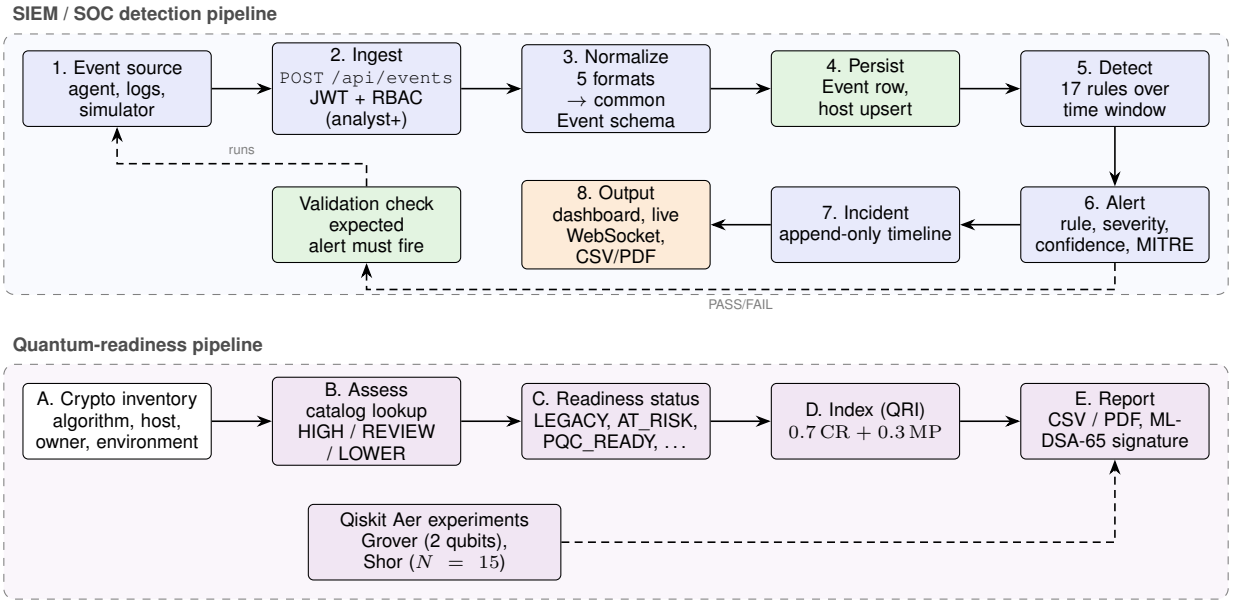


Fig. 2. Data pipelines. Top: ingestion, detection, alerting and incident handling, with the validation check that confirms a simulated scenario produced its expected alert. Bottom: cryptographic inventory to Quantum Readiness Index and signed report.

TABLE IV
RELATIONAL DATA MODEL (FROM THE MIGRATION SCRIPTS AND ORM MODELS).

Table	Purpose	Principal columns and relations
`users`	Accounts and roles	email (unique), Argon2 hash, role (admin/analyst/viewer), active flag
`events`	Normalized SIEM events	timestamp, source, host, event type, severity, user, source/destination address, port, message, raw JSON; indexed on seven columns
`detection_rules`	Rule metadata	rule identifier (unique), category, severity, MITRE technique, enabled flag
`alerts`	Rule firings	rule, severity, confidence, status (OPEN/ACKNOWLEDGED/CLOSED), evidence event IDs, host, source address, acknowledging user
`incidents`	Investigations	title, status (six values), severity, assigned analyst, resolution, timestamps
`incident_alerts`	Incident–alert links	foreign keys to incidents and alerts
`incident_timeline`	Append-only trail	entry type (note, status change, alert linked), author, content, time
`hosts`	Asset directory	hostname, address, environment, owner, first/last seen; upserted on ingest
`audit_logs`	Security-relevant actions	actor, action, target, JSON detail, time
`simulation_runs`	Simulator history	scenario, triggering user, events generated, expected rule, detection confirmed
`crypto_assets`	Cryptographic inventory	algorithm, system, purpose, key size, service, host, port, protocol, certificate info, environment, owner, readiness status
`quantum_assessments`	Assessment results	optional asset link, crypto type, threat, risk class, migration status, recommendation, readiness status
`quantum_experiments`	Qiskit runs	algorithm, backend, qubits, shots, duration, measurement counts, dominant result
`readiness_scores`	QRI snapshots	component scores, weights, index, label

A viewer token that reaches a mutating endpoint receives HTTP 403 and not a degraded response; this is covered by tests for events, alerts, incidents and the simulator. Login is substantially limited to ten attempts per minute per client address applying an in-memory, per-process limiter; this is satisfactory for a person worker, but would not be shared across the workers.

TABLE V
ROLE-BASED ACCESS CONTROL.

Role	Read	Ingest / simulate / manage	Users
viewer	yes	no	no
analyst	yes	yes	no
admin	yes	yes	future

G. Application programming interface

The backend exposes 16 route modules (admin, agent, alerts, auth, dashboard, endpoints, events, exports, incidents, IOCs, live, PQC, quantum, reports, rules, simulator). Table VII lists the principal endpoints that the evaluation touches. Deviations use a uniform `{"detail": ...}` body with 400 for malformed ingestion, 401 for authentication failures, 403 for part failures, 404 for missing resources, 422 for schema validation, 429 for login throttling and 500 for unhandled discrepancies. The frontend has 21 pages; an endpoint agent

TABLE VI
SECURITY CONTROLS AND THE THREAT EACH ADDRESSES.

Control	Threat addressed
JWT (HS256), secret from environment	Unauthenticated access; secrets in source
Argon2 password hashing	Credential disclosure from the database
Login rate limit, 10 per minute	Credential stuffing against the dashboard
Fixed ingestion formats, 400 on malformed	Unrecognized or partial payloads stored
ORM-only queries, Pydantic bodies	SQL injection
Audit log of logins, alert and incident changes, runs	Unaccountable privileged actions
Simulator posts only to its own API	Real attack traffic from a teaching tool
Documentation-range addresses (203.0.113.0/24)	Confusion with real hosts
Configurable CORS origins	Open cross-origin access

TABLE VII
PRINCIPAL API ENDPOINTS (A+: ANALYST OR ADMIN).

Method	Path	Auth	Function
POST	`/api/auth/login`	none	Issue token
POST	`/api/events[/batch]`	A+	Normalize, store, detect
GET	`/api/events`	any	Filtered event list
PATCH	`/api/alerts/{id}`	A+	Acknowledge or close
POST	`/api/incidents`	A+	Open incident from alerts
GET	`/api/dashboard/summary`	any	Combined KPIs
POST	`/api/simulator/run`	A+	Run a scenario
POST	`/api/quantum/assess`	A+	Classify an algorithm
GET	`/api/quantum/readiness`	A+	Compute and store QRI
GET	`/api/quantum/grover|shor`	A+	Run Qiskit demo
GET	`/api/reports/readiness/pdf`	any	Signed readiness report

(about 600 non-blank lines) enrolls with a single-use token, buffers events while offline and is, in effect, rejected promptly after revocation, which the root test suite checks against a real server.

IV. DETECTION ENGINE AND RULE CATALOGUE

A. Event normalization

Whatever the source, an event is reduced to one schema: timestamp, source, host, event type, severity, user, source address, destination address and port, message and the original payload. Table VIII lists the five accepted formats. Unknown formats, missing required fields and invalid severities, actions or Windows event identifiers raise a normalization error that is returned as HTTP 400, so nothing is partially stored. The Windows mapping covers only a documented subset of Security event identifiers and is not a claim of full Event Log coverage. The simulator format is normalized identically to the generic one, so detection rules never special-case simulated events; this is what allows the simulator to test the real path.

TABLE VIII
ACCEPTED INGESTION FORMATS.

Format	Required fields	Mapping
generic	host, event type	Pass-through; severity defaults to medium, time to now
linux_auth	host, action	failed/invalid/accepted password or key, sudo, session → event type and severity
windows_security	Computer, EventID	4625, 4624, 4672, 4720, 4732 → failure, success, privilege use, account created, escalation
network	host, source address	Default type connection attempt; carries destination and protocol
simulator	as generic	Identical to generic

B. Rule catalogue

17 rules in five modules (authentica- are encompassed by the engine tion, network, web and endpoint, with any second center on threat intelligence). Each module exports static metadata and the `evaluate` function, and all modules on every ingested batch are run by the engine, seeds the rule table idempotently, and suppresses alerts that are repeating Within five minutes, the same rule and host or source tackle. Table IX lists every rule with its trigger, severity and ATT&CK method as recorded in the code. Rules fall into two kinds: *threshold* rules are specifically counting events per source address or host in a sliding window, and *typed-event* rules fire when an event already carries a particular type. Threshold rules are, in essence, the ones with behaviour worth testing at their boundaries (Section VII); typed-event rules rely on the upstream producer, here the simulator or agent, to classify the event.

C. Attack simulator

Descriptive telemetry is only generated by the simulator. Its scenarios call the application's own ingestion path; they never open one socket to one more host, never issue a request to a third party and never run an operating-system tool. The port-scan scenario posts network events shaped like a scan, and the ransomware-like situation posts file-activity events outlining a mass rename, without yielding, renaming or encrypting any file. Running a case needs an analyst token, creates a simulation record and writes an audit entry. Table X recapitulates the 11 situations, one per projected rule family. The returned `detection_confirmed` flag is the signal of interest, including it is true only if the declared rule actually raised an alert for that run. The situation for the threat-intelligence rule applies a manifestly labelled test indicator, and the brute-force scenario is marked as a lab procedure.

D. Assessment catalog

21 normalized algorithm is held by the quantum layer's catalog identifiers, namely seven HIGH, two REVIEW and twelve LOWER. Table XI is listing them. Each entry carries a type, a category, a migration status and a reference string that is always either a theoretical citation or a NIST document;

TABLE IX
DETECTION RULE CATALOGUE (17 RULES). TRIGGERS ARE FROM THE DETECTION-RULE REFERENCE AND WERE CHECKED AT THEIR BOUNDARIES (E3); THE TECHNIQUE COLUMN IS THE MAPPING IN THE CODE, WITH ITS QUALITY (D: DIRECT, I: INDICATIVE, –: NONE).

Rule	Severity	Trigger	Technique	Q
auth.repeated_failed_logins	medium	≥ 5 authentication failures, same source address, within 10 min	T1110	D
auth.brute_force_burst	high	≥ 10 authentication failures, same source address, within 5 min	T1110	D
auth.successful_login_after_failures	high	Success preceded by ≥ 3 failures from the same address within 10 min	T1078	I
auth.privilege_related_activity	medium	Any privilege-use event (high if privilege escalation)	T1078	I
network.repeated_connection_attempts	medium	≥ 20 connection attempts, same source, within 5 min	–	–
network.synthetic_port_scan	high	Connection attempts to ≥ 8 distinct ports from one source within 2 min	T1046	D
network.unexpected_service_access	medium	Event typed unexpected service access	–	–
web.repeated_api_auth_failures	medium	≥ 5 API authentication failures, same source, within 5 min	T1110	D
web.abnormal_request_rate	medium	≥ 50 API requests, same source, within 1 min	T1499	I
web.suspicious_request_pattern	high	Event typed suspicious web request	T1190	I
endpoint.suspicious_process	medium	Event typed suspicious process	–	–
endpoint.privilege_escalation_indicator	high	Event typed privilege-escalation indicator	–	–
endpoint.persistence_indicator	high	Event typed persistence indicator	T1547	I
endpoint.suspicious_file_activity	medium	Single event typed suspicious file activity	–	–
endpoint.ransomware_like_file_activity_burst	critical	≥ 5 suspicious file-activity events, same host, within 2 min	T1486	I
endpoint.data_access_anomaly	medium	Event typed data-access anomaly	T1005	I
threat_intel.ioc_match	medium	Event field matches a stored (test) indicator	–	–

TABLE X
SIMULATOR SCENARIOS AND GENERATED EVENTS.

Scenario	Events	Expected rule
auth failure burst	12	brute_force_burst
repeated login attempts	6	repeated_failed_logins
success after failures	5	successful_login_after_failures
synthetic port scan	12	synthetic_port_scan
suspicious web request	1	suspicious_request_pattern
privilege escalation	1	privilege_escalation_indicator
suspicious process	1	suspicious_process
persistence indicator	1	persistence_indicator
data access anomaly	1	data_access_anomaly
ransomware-like activity	6	ransomware_like_file_burst
test indicator match	1	ioc_match

the engine returns an unambiguous UNKNOWN for anything absent, in preference to a invented answer. In this broader context, the catalog is predominantly hand-curated and is uniformly not an formal NIST security-level assignment or a certified audit result.

E. What Q-SOC itself employs

The store classifies its own cryptography, which is a valuable consistency check for a readiness tool. Report and export signatures apply ML-DSA-65 with a context string, and the round trip is tested. Argon2id are employed by passwords, and session tokens apply an HMAC-SHA256 JWT, which is a symmetric construction outside Shor's reach but not a post-quantum signature. Under these conditions, the lab TLS certificate uses RSA-3072 and is therefore reported as LEGACY, because post-quantum certificates are not yet broadly supported by browsers, agents or certificate authorities. The hybrid X25519MLKEM768 key exchange in nginx was showed by a handshake test against a local server, but the nginx and Docker side was not executed in the build environment. Because, the RSA certificate stays flagged, the platform is deliberately not reported as fully quantum ready and its own index remains below 100.

V. DEMONSTRATION RUN AND INTERFACE OUTPUTS

To reveal what an analyst sees without counting on interface images, this section reports the contents of the application's dashboard, alert, validation, inventory, readiness and experiment pages as tables and graphs, created by script from the application's own API (`scripts/make_demo_figs.py`, data in `data/demo_data.json`). The instance was a development-mode rollout (FastAPI with SQLite, Vite-served frontend), populated only through the application's own interfaces, namely the 11 simulator scenarios were run, one incident was opened from three of the consequent alerts, ten cryptographic assets were added to the two seeded ones and assessed, and the Grover and Shor trials were, in practice, executed. All data are synthetic and all tackles are widely in the documentation range.

Dashboard. Table XII is listing the indicators this dashboard shows after the run. Fig. 3 is distributing the 11 alerts by severity and rule category and the events per host. Validation was passed by all 11 cases and none failed, and the one critical alert came from the ransomware-like file-activity scenario.

Alerts and detection validation. Table XIII lays out, for each individual simulator scenario, the rule it is predominantly anticipated to trigger, the projected severity and ATT&CK technique, the confidence of the ensuing alert, the number of events the situation produced and the outcome. The outcome column is typically obtained by checking the alert raised by the application with the scenario's declared expectation; it is agreeing with the application's own validation tally of 11 passed and 0 failed. Three scenarios (privilege escalation, suspicious process and the threat-No ATT&CK technique tag is had in this by intelligence match) run. Alerts are starting as open and the analyst can admit them or link them to one more incident, which in this run carried any timeline of the

TABLE XI
ASSESSMENT CATALOG (21 ENTRIES).

Algorithm	Purpose	Risk	Readiness	Basis
RSA-2048 / 3072 / 4096	key establishment	HIGH	LEGACY	Shor; NIST PQC guidance
ECDH, DH-2048	key establishment	HIGH	LEGACY	Shor; NIST IR 8547 draft transition plan
ECDSA, ECC	signature	HIGH	LEGACY	Shor (extended to elliptic curves)
AES-128	symmetric	REVIEW	AT_RISK	Grover; reduced strength margin
SHA-256	hash	REVIEW	AT_RISK	Grover; reduced preimage margin
AES-256, HMAC-SHA256	symmetric / MAC	LOWER	PARTIALLY_READY	Grover; adequate margin
SHA-384, Argon2id	hash / KDF	LOWER	PARTIALLY_READY	Grover; RFC 9106 for Argon2id
ML-KEM-512 / 768 / 1024	key establishment	LOWER	PQC_READY	NIST FIPS 203
ML-DSA-44 / 65 / 87	signature	LOWER	PQC_READY	NIST FIPS 204
SLH-DSA	signature	LOWER	PQC_READY	NIST FIPS 205
X25519MLKEM768	hybrid key exchange	LOWER	PQC_READY	FIPS 203 component; IETF hybrid draft

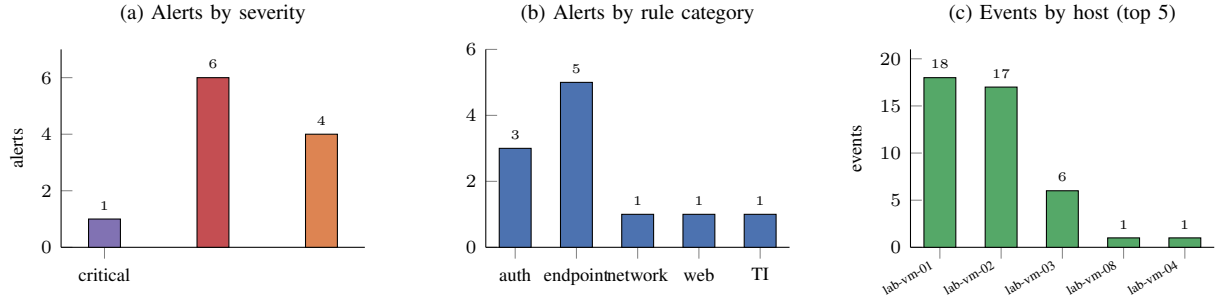


Fig. 3. Demonstration run (47 events, 11 alerts): (a) alerts by severity, (b) alerts by detection-rule category (TI: threat intelligence), (c) events by host.

TABLE XII
DASHBOARD INDICATORS AFTER THE DEMONSTRATION RUN.

Indicator	Value
Total events	47
Alerts (critical / high / medium)	11 (1 / 6 / 4)
Active incidents	1
Validation runs passed / failed	11 / 0
Threat-intel indicators (active / test)	5 / 5
Threat-intel matches	1
Crypto inventory assets / assessments	12 / 10
Quantum-vulnerable algorithms (assessed)	4
Quantum Readiness Index	54
Migration readiness	40.0%
Qiskit experiments run	2

creation note, the status are changing to STUDYING and one analyst note.

Quantum security pages. Table XIV is listing the cryptographic inventory with the risk classification from the assessment engine and the formulated readiness status. Fig. 4 recapitulates the three QRI components, the readiness-status distribution of the 12 assets and the risk classification of the 10 assessments. With component scores of 60, 40 and 60 and the default weights, the index is $0.4{\cdot}60{+}0.3{\cdot}40{+}0.3{\cdot}60 = 54$. Four assessed algorithms (RSA-2048, RSA-3072, ECDSA and ECDH) are classified HIGH risk, AES-128 and SHA-256 REVIEW, AES-256 and ML-DSA-65 LOWER, and 3DES and SHA-1 UNKNOWN because the catalog enters for them; Section VII is explaining why that last case matters for the index. Two seeded assets are, in essence, pointed to as not assessed.

Trials page. Fig. 5 is revealing the measurement counts recorded by the two Qiskit experiments. The 2-qubit Grover The marked state `11` was returned in all 1024 shots on the by search noiseless simulator. The Shor demonstration for $N = 15$, $a = 7$ returned four roughly equiprobable outcomes (0, 4, 8, 12 of 16), coherent with an order $r = 4$, from which $\gcd(7^2 \pm 1, 15)$ gives the factors 3 and 5. Both are, in effect, toy instances on a standard simulator.

VI. EVALUATION METHOD

All experiments import the unmodified backend and write raw CSV/JSON results, from which the figures and the signing table are generated. Host: Linux VM, 2 vCPU (i7-1255U), 3.9 GB RAM, Python 3.13.16, Qiskit 2.5.2, Aer 0.17.2, SQLite; Docker, PostgreSQL and nginx were not used. Random seeds are recorded in the result files (QRI inventories: seed 20261013). Experiments (Table XV): R1 (QRI properties, 2000 random inventories of 3–30 assets), E1 (HTTP ingestion on a real uvicorn server, 11,800 events), E2 (scenario validation, 20 repetitions $\times$ 11 scenarios, fresh database each), E3 (24 boundary cases), E5 (shared-address noise, synthetic), E9 (37 malformed-input cases), Q1–Q3 (Qiskit) and S1 (signing cost, 30 repetitions).

Each experiment writes its raw rows to CSV or JSON together with the environment record (operating system, Python and package versions, repository commit d7ad437 and a flag stating that the working tree equalled HEAD), so the tables and

TABLE XIII
SIMULATOR SCENARIOS, EXPECTED DETECTIONS AND OBSERVED ALERTS IN THE DEMONSTRATION RUN (EVENTS SUM TO 47).

Scenario	Expected rule	Severity	Conf.	ATT&CK	Events	Result
auth_failure_burst	`auth.brute_force_burst`	high	0.85	T1110	12	PASS
repeated_login_attempts	`auth.repeated_failed_logins`	medium	0.70	T1110	6	PASS
successful_login_after_failures	`auth.successful_login_after_failures`	high	0.80	T1078	5	PASS
synthetic_port_scan	`network.synthetic_port_scan`	high	0.80	T1046	12	PASS
suspicious_web_request	`web.suspicious_request_pattern`	high	0.75	T1190	1	PASS
privilege_escalation_indicator	`endpoint.privilege_escalation_indicator`	high	0.80	none	1	PASS
suspicious_process_event	`endpoint.suspicious_process`	medium	0.70	none	1	PASS
persistence_indicator	`endpoint.persistence_indicator`	high	0.80	T1547	1	PASS
data_access_anomaly	`endpoint.data_access_anomaly`	medium	0.65	T1005	1	PASS
ransomware_like_file_activity	`endpoint.ransomware_like_file_activity_burst`	critical	0.90	T1486	6	PASS
ioc_test_indicator_match	`threat_intel.ioc_match`	medium	1.00	none	1	PASS

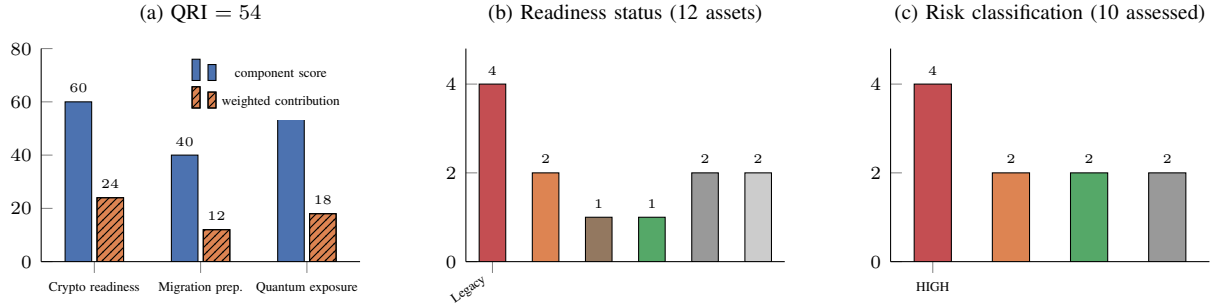


Fig. 4. Quantum-readiness outputs of the demonstration run: (a) QRI component scores and weighted contributions (QRI = 54), (b) readiness status of the 12 inventory assets, (c) risk classification of the 10 assessments.

TABLE XIV
CRYPTOGRAPHIC INVENTORY OF THE DEMONSTRATION RUN.

Algorithm	Host	Risk class	Readiness status
ML-DSA-65	q-soc	LOWER	PQC_READY
3DES	pay-gw	UNKNOWN	UNKNOWN
SHA-1	legacy-01	UNKNOWN	UNKNOWN
SHA-256	ci-01	REVIEW	AT_RISK
AES-128	backup-01	REVIEW	AT_RISK
AES-256	db-01	LOWER	PARTIALLY_READY
ECDH-P256	web-01	HIGH	LEGACY
RSA-3072	vpn.example.internal	HIGH	LEGACY
ECDSA-P256	build-01	HIGH	LEGACY
RSA-2048	api.example.internal	HIGH	LEGACY
AES-256	seeded demo asset	not assessed	NOT_ASSESSED
RSA-2048	portal.lab.local	not assessed	NOT_ASSESSED

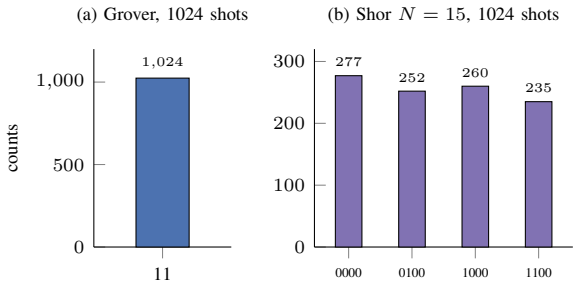


Fig. 5. Measurement counts recorded by the Qiskit experiments page (local Aer simulator).

figures can be regenerated without re-running the workload. Four design principles guided the harness. (i) The system under test is the unmodified backend, imported or started as a server, so there is no re-implementation to drift from the product. (ii) Every positive check has a negative control: the scenario validation is repeated with the expected rule disabled, which must fail. (iii) Expectations are written from the documented specification before the run, and a deviation is reported as a finding rather than adjusted. (iv) Synthetic parameters (for example the failure probability in E5) are stated, and results from them are described as properties of the thresholds, not as field measurements.

VII. RESULTS

A. Detection validation (E2, E3)

All 220 simulator runs passed; with the anticipated rule disabled, 33 of 33 runs failed, so the check is sensitive to the rule rather than empty. In-process median latency was roughly 20–36 ms (p95 30–71 ms). Over HTTP on the loaded database all 11 situations still passed (52–73 ms). The 24 boundary cases conformed to the rule specification. Three behaviours matter for users. In streaming mode `repeated_failed_logins` fires at event 5 and `brute_force_burst` at event 10, so Within any batch, “supersedes” is holding only. Ten failures spaced one hour apart (low-and-slow) and 20 failures from 20 sources (distributed) created no alert, the second by-design

TABLE XV
EXPERIMENT DESIGN.

ID	Question	Workload	Metrics	Runs
E1	Throughput, latency	HTTP ingestion to a live uvicorn server; heartbeat, authentication success and failure events; batch size $B \in \{1, 10, 100, 500\}$; read endpoints on 11,800 stored events	Events/s, request latency, alerts raised, read latency	100/30/10/5 per B; 30 per read
E2	Scenario validation	11 scenarios, fresh database per run; same with the expected rule disabled; immediate re-run	Pass/fail, latency, severity match	$20 \times 11 + 33 + 11$
E3	Rule boundaries	Event sequences at, just below and just above each threshold; window edges; evasion; deduplication	Fire/no-fire vs. specification	24 cases
E5	Shared-address noise	$U \in \{25, 100, 400, 1000\}$ benign users behind one address, six attempts per day, failure probability $p \leq 0.10$	Alerts per day by rule	5 seeds per cell
E9	Input robustness	30 malformed or unusual payloads plus 7 protocol-level checks	HTTP status, server failure, conformance	37 cases
R1	QRI properties	Random inventories of 3–30 assets drawn with replacement from the catalog; 66 weight vectors	Equality of components, rank correlation	2000 inventories
Q1–Q3	Quantum demonstrations	Grover on 2 qubits, ideal and under depolarizing noise; Shor for $N = 15$	Success probability, depth, gate counts	80 / 20 per noise level / 50
S1	Signing cost	ML-DSA-65, ECDSA P-256, RSA-3072-PSS; 1 KiB to 10 MiB	Sign and verify time, sizes	30

limit of tackle-keyed windows. Finally, six cases without one source address failed when re-run instantly inside the deduplication window; while the five with tackles passed.

Table XVI and Fig. 6 are giving the per-scenario view. Latency is, in this setting, mostly lowest for the single-event typed rules and higher for the multi-event threshold scenarios, as projected from the number of events normalized and queried. The immediate re-run column is separating the deduplication finding, namely the six situations that fail on a second run are those whose events carry no source address, so the suppression key can't distinguish the second run from the first. An analyst re-running such a situation would therefore see a failure that reflects the suppression policy, not a defect in the rule.

The boundary table (Table XVII) is revealing that each individual threshold is executed as stipulated: the rule is, in effect, silent one event below the limit and fires exactly at it. The window cases are validating that the span, and not the count alone, is deciding, and the evasion rows are recording what the specification can't catch. Two further details are, in effect, worth stating. In cases B02 and B17 the higher rule and an alert for the was raised by its lower-threshold companion both same sequence, so two alerts exist for one behaviour; in B01 and B16 a lower-severity companion rule raised an alert when the higher rule did not, which is the intended escalation ladder.

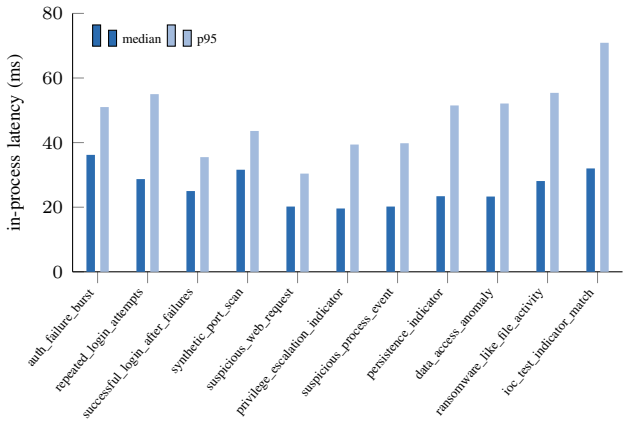


Fig. 6. In-process scenario latency (median and p95), 20 runs per scenario (E2).

B. Shared-address noise (E5)

With U users behind one tackle attacking per day, no alerts emerged at $U = 25$ for any failure probability tested ($p \leq 0.10$). At $U = 400$, mean alerts per day were 0.8, 11.2, 31.8 and 76.4 for $p = 0.01, 0.03, 0.05, 0.10$; at $U = 1000$ they were 14.0, 68.2, 113.2 and 140.6 (Fig. 7). Most came

TABLE XVI
PER-SCENARIO RESULTS (E2): 20 RUNS EACH, IN-PROCESS LATENCY IN MS, NEGATIVE CONTROL (RULE DISABLED, 3 RUNS EACH) AND IMMEDIATE RE-RUN INSIDE THE FIVE-MINUTE DEDUPLICATION WINDOW.

Scenario	Events	Median	p95	Pass (of 20)	Control (rule off)	Immediate re-run
auth_failure_burst	12	36.2	51.0	20	fail 3/3	pass
repeated_login_attempts	6	28.7	55.0	20	fail 3/3	pass
successful_login_after_failures	5	25.0	35.5	20	fail 3/3	pass
synthetic_port_scan	12	31.6	43.6	20	fail 3/3	pass
suspicious_web_request	1	20.2	30.4	20	fail 3/3	pass
privilege_escalation_indicator	1	19.6	39.4	20	fail 3/3	**fail**
suspicious_process_event	1	20.2	39.8	20	fail 3/3	**fail**
persistence_indicator	1	23.4	51.5	20	fail 3/3	**fail**
data_access_anomaly	1	23.3	52.1	20	fail 3/3	**fail**
ransomware_like_file_activity	6	28.1	55.4	20	fail 3/3	**fail**
ioc_test_indicator_match	1	32.0	70.9	20	fail 3/3	**fail**

TABLE XVII
BOUNDARY, WINDOW, EVASION AND DEDUPLICATION CASES (E3). "FIRES" IS THE OBSERVED OUTCOME; ALL 24 CASES CONFORMED TO THE RULE SPECIFICATION.

ID	Rule under test	Input	Expected	Fires
B01/B02	brute_force_burst	9 / 10 failed logins, 1 s apart	no / yes	no / yes (event 10)
B03/B04	repeated_failed_logins	4 / 5 failed logins, 10 s apart	no / yes	no / yes (event 5)
B05/B06	success_after_failures	2 / 3 failures then a success, same source	no / yes	no / yes
B07/B08	synthetic_port_scan	7 / 8 distinct ports, 5 s apart	no / yes	no / yes
B09	synthetic_port_scan	8 attempts to the same port	no	no
B10/B11	repeated_connection_attempts	19 / 20 attempts, same port	no / yes	no / yes
B12/B13	repeated_api_auth_failures	4 / 5 API failures, 5 s apart	no / yes	no / yes
B14/B15	abnormal_request_rate	49 / 50 requests in 60 s	no / yes	no / yes
B16/B17	ransomware_like_burst	4 / 5 file events, one host, 10 s apart	no / yes	no / yes
W01	brute_force_burst	10 failures, 40 s apart (span 360 s $>$ 300 s)	no	no
W02	repeated_failed_logins	same 10 failures, 600 s window	yes	yes (event 5)
W03	repeated_failed_logins	5 failures, 160 s apart (span 640 s)	no	no
E01	repeated_failed_logins	low-and-slow: 10 failures, 1 h apart	(evasion)	no alert
E02/E03	repeated_failed / brute_force	distributed: 20 failures from 20 sources	(evasion)	no alert
D01	brute_force_burst	two 12-failure bursts, 2 min apart	one alert	one alert

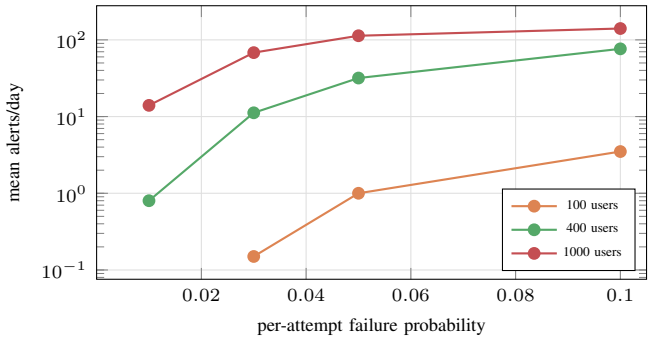


Fig. 7. Alerts per day from benign users sharing one address (synthetic; 5 seeds per cell). 25 users produced no alerts at any tested p.

from `successful_login_after_failures`. This is a attribute of the thresholds under our stated model, not a measured false-positive rate.

C. Throughput and latency (E1)

Fig. 8 is revealing throughput scaling with batch size; for heartbeats the median rose from 34.8 events/s ($B = 1$) to 1525.5 ($B = 500$), with median request latency 28.7 ms and 327.8 ms. Per-request latency at $B = 10$ was about 100 ms, non-monotonic against neighbours; we did not study the cause. At 11,800 stored events, median read latencies were 1.3 ms (health), 7.4 ms (events list), 3.7 ms (alerts list), 29.0 ms (dashboard summary) and 269.6 ms (dashboard timeline). Detection runs inside the request, so the figures bundle ingestion and detection. They are outlining one laptop-class VM and are notably not capacity claims.

Table XVIII and Fig. 9 show that the batching effect holds across the event types, but its size differs: heartbeats reach about 1526 events/s at $B = 500$, authentication successes 1218 and authentication failures with separate sources 940. Failure events cost more per event, because each one also is invoking the authentication-rule queries. The variance is broadly large at small batch sizes (to reveal, heartbeat throughput at $B = 1$ ranged from 8 to 66 events/s across 100 requests), so single-event medians should be read with that spread in mind.

The read paths in Fig. 10 are separating into two groups. Indexed list endpoints and the health check stay in single-digit milliseconds, while the dashboard summary and chiefly the hourly timeline aggregate over the full table, taking 29 and 270 ms. The timeline is the obvious first target if the stored

TABLE XVIII
MEDIAN INGESTION THROUGHPUT (EVENTS/S) AND REQUEST LATENCY (MS) BY EVENT TYPE AND BATCH SIZE, HTTP, SQLITE (E1). NO ALERTS WERE RAISED IN ANY OF THESE WORKLOADS.

Event type	B=1	B=10	B=100	B=500
heartbeat (events/s)	34.8	97.3	741.5	1525.5
latency (ms)	28.7	102.8	134.9	327.8
auth success (events/s)	25.7	104.8	797.5	1218.2
latency (ms)	39.0	95.5	125.4	410.4
auth failure, distinct sources (events/s)	48.9	343.1	785.2	939.8
latency (ms)	20.5	29.2	127.4	532.0

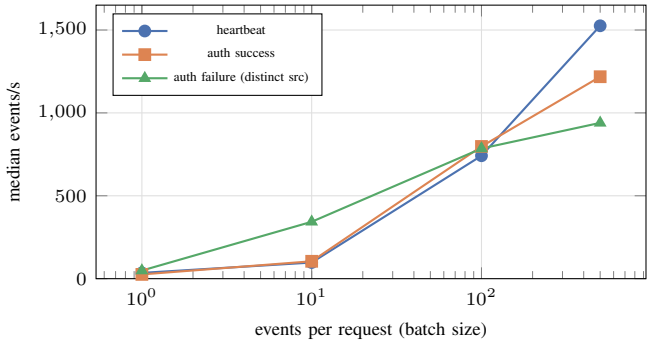


Fig. 8. Median ingestion throughput vs. batch size over HTTP (SQLite, single process).

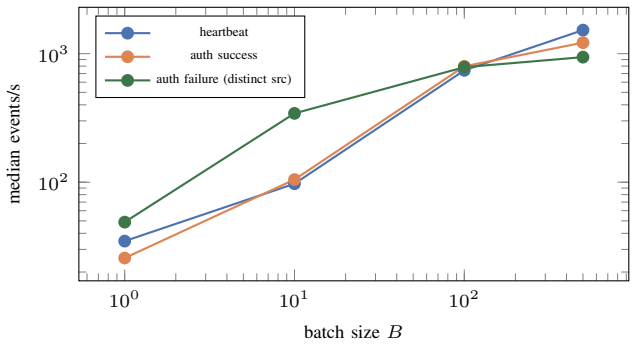


Fig. 9. Median throughput by event type and batch size (log–log).

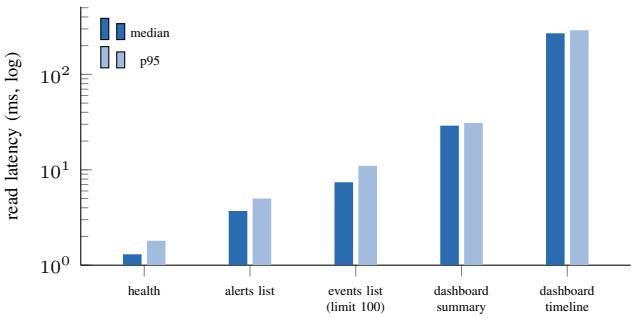


Fig. 10. Read-path latency on 11,800 stored events (30 requests each).

volume is growing, as any illustration through one more pre-aggregated bucket table.

D. Input sturdiness (E9)

Of 37 cases, 27 strict expectations were met, namely format and field errors returned 400, wrong payload shapes 422, unauthenticated requests 401 and viewer writes 403; batch writes were atomic, secrets were, in practice, redacted and SQL-injection strings were stored inertly. Six cases caused server failures (a list-valued host, one dict-valued host, a non-numeric port, an oversized port, a dict-valued username and an integer message). The service stayed healthy afterwards. The syntactically invalid tackle 999.1.1.1 was accepted, so no address validation applies. We did not adjust the application; type validation at the schema layer is the obvious solve.

The failure cases in Table XIX are sharing one cause, like the nor- A value of the wrong type is accepted by malizer and a later step (a database insert or string operation) fails. Of the 37 cases, 27 met their strict expectation; 10 were exploratory probes without one. Six of these ten caused a server failure, three returned HTTP 500 and three ended with the connection closed without a response, while the other four were accepted and stored. None left the service unhealthy, and the post-test health check passed. Because the failures are reached by an authenticated analyst and not by an anonymous caller, the exposure is bounded, but it is still the most direct improvement available.

E. QRI properties (R1)

On 2000 random inventories, CR equalled QE in every case and also equalled MP, and the default QRI equalled $0.7\,\mathrm{CR} + 0.3\,\mathrm{MP}$ exactly. Spearman correlation against the default ranking was 1.0 for all 66 alternative weight vectors, so the weights change magnitudes but not orderings on this generator. Unclassified algorithms are the notable edge: CR and QE count them as non-HIGH, whereas MP treats them as not prepared. Ten unclassified assets yield $\mathrm{CR} = 100$, $\mathrm{MP} = 0$, $\mathrm{QRI} = 70$, while ten RSA-2048 assets yield $\mathrm{QRI} = 0$. The CR$-$MP gap grows linearly with the unclassified fraction (Fig. 11). Reference inventories scored 40 (legacy-heavy), 60 (mixed transition) and 100 (PQ-first). Because CR, MP and QE coincide on catalog-only inventories, the index offers no independent information from its three components there. The demonstration inventory (Table XIV) contains two UNKNOWN algorithms, so this behaviour is not hypothetical.

Tables XX and XXI make the index behaviour concrete. Over 2000 random inventories the mean QRI was 66.9 (standard deviation 14.2, 95th percentile 87.5). Adding unclassified assets raises the mean index, from 66.8 with none to 70.0 with all assets unclassified, which is the opposite of what a cautious tool should do when it knows less about an inventory. This directly motivates the recommendation to report unclassified assets as their own bucket.

F. Quantum demonstrations (Q1–Q3)

Ideal Grover runs succeeded with probability 1.0 (80 runs). Under depolarizing noise ($p_2 = 10p_1$, 20 repetitions of 1024 shots) mean success fell from 1.0000 ($p_1 = 0$) to 0.9648 at 10^{-3}, 0.7114 at 10^{-2} and 0.2979 at 5×10^{-2} (Fig. 12). In basis $\{rz, sx, x, cx\}$ the circuit has depth 26. The Shor toy

TABLE XIX
ROBUSTNESS CASES BY CLASS (E9).

Class	Cases	Observed outcome
Unknown format, missing field, bad value	11	400 (as specified)
Wrong payload shape, invalid JSON	4	422
Unauthenticated / viewer write	2	401 / 403
Empty batch, batch atomicity	2	400, nothing stored
Oversize, nesting, Unicode, injection, valid address	6	accepted (201) and stored inertly
Secret redaction, post-test health	2	as specified
Control characters, long string, invalid address	3	accepted (201); no strict expectation
Wrong field type (list, dict, text/large port, integer)	7	1 accepted, 3 HTTP 500, 3 dropped

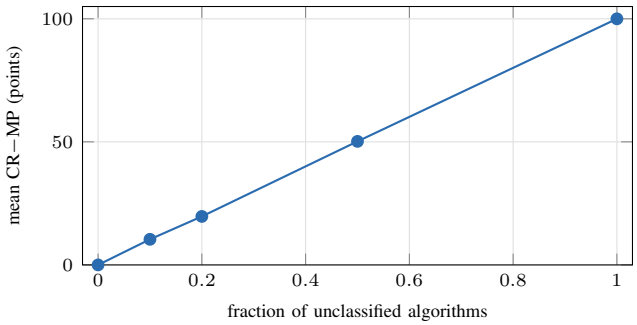


Fig. 11. Mean CR−MP gap vs. fraction of unclassified algorithms (R1).

($N = 15$, $a = 7$, 8 qubits) recovered $\{3, 5\}$ in 50/50 runs at 1024 shots; its depth is 189 in Aer's gate set and 1700 in the restricted basis (888 CX). Under these conditions, all these are revealing one unimportant instance and say nothing about the feasibility of attacks on real keys, for which see [9].

Table XXII places the two circuits side by side. The Shor outcomes are widely close to uniform over the four values that are multiples of $16/r$, as theory projects for $r = 4$, and any further chi-square statistic of about 1.3 on three degrees of freedom is uniform with equal probabilities. The jump from the depth of 26 to 1700 for the smallest non-trivial factoring instance reveals, without any claim about real keys, why resource approximations for 2048-bit moduli [9] are regularly so large.

G. Signing cost (S1)

ML-DSA-65 signatures are 3309 B with 1952 B public keys, against 70 B (ECDSA P-256, DER) and 384 B (RSA-3072) signatures. For report-sized payloads the extra time is milliseconds (Table XXIII).

VIII. EVIDENCE STATUS AND SHORTCOMINGS

Documentation and code are disagreeing in places – the detection-rule document and README list older MITRE mappings or 16 rules (code: 17), one document says 10 situations (code: 11), and 172 backend tests are reported by documents (we gathered 193). We follow the code. The demonstration run of Section V applied a development-mode instance (SQLite, Vite server), not the Docker/nginx/PostgreSQL deployment, which we did not run. Further shortcomings – all event data are synthetic; one host; SQLite only; no comparison with another SOC platform; no user study. The 19 cited works are, in practice, journal articles; each was checked against Crossref for metadata and against SCImago (SJR 2025) for quartile, and each individual venue is substantially ranked Q1 as its best quartile. One venue, Sensors, is reliably Q1 in electrical and electronic engineering but Q2 in information systems; a candidate magazine (IEEE Security & Privacy) was excluded because its Q1 ranking is only in the Law category. Quartiles are, in effect, current rankings, not those at publication, and are commonly listed in `references_verification.csv`.

TABLE XX
REFERENCE INVENTORIES AND EDGE CASES FOR THE QRI (R1).

Inventory (10 assets)	CR	MP	QE	QRI
Legacy-heavy (4 RSA-2048, 2 ECDSA, 2 AES-128, 2 SHA-256)	40	40	40	40
Mixed (2 RSA-2048, 2 ECDH, 2 hybrid KEM, 2 ML-DSA-65, 2 AES-256)	60	60	60	60
PQ-first (hybrid and ML-KEM, ML-DSA-65, AES-256)	100	100	100	100
Ten RSA-2048 assets	0	0	0	0
Ten unclassified assets	100	0	100	70

TABLE XXI
EFFECT OF UNCLASSIFIED ALGORITHMS ON THE INDEX (R1; MEAN OVER RANDOM INVENTORIES).

Fraction unclassified	Mean CR−MP	Mean QRI
0.0	0.0	66.8
0.1	10.4	67.4
0.2	19.7	67.4
0.5	50.2	69.2
1.0	100.0	70.0

A. Threats to validity

Construct validity. A case is passing when the declared rule fires, so the validation measures agreement between a generator and a rule written by the same authors. It lays out that the pipeline is, in this setting, wired correctly and that the negative control is sensitive; it does not reveal that the rules detect real opponents. *Internal validity.* Timing was, in essence, taken on a two-core virtual machine whose host load we did not control, which is, in practice, explaining some of the spread (as an illustration the 62.8 ms standard departure of the threat-intelligence situation against a 32 ms median). Seeds are fixed and recorded. *External validity.* All events are, in essence, synthetic, the database is SQLite, and there is one host; PostgreSQL, multiple workers, TLS termination and real agents may, in essence, change both throughput and failure modes. *Conclusion validity.* Counts such as 20 repetitions per case tightly bound on a deterministic pipeline but say little about rare faults; the noise experiment E5 is a model with stated factors. *Metric validity.* The QRI is a suggested research

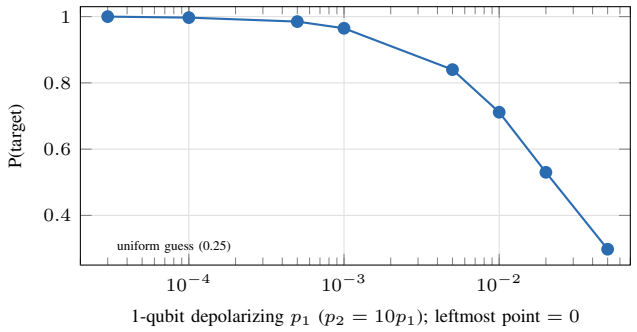


Fig. 12. Grover success vs. depolarizing noise.

TABLE XXII
QUANTUM CIRCUIT STATISTICS (Q2, Q3).

Quantity	Grover (2 qubits)	Shor ($N = 15$, $a = 7$)
Logical qubits	2	8 (4 counting, 4 work)
Depth, Aer gate set	–	189
Depth, basis $\{rz, sx, x, cx\}$	26	1700
CX gates, basis	2	888
Other gates	20 rz, 10 sx, 4 x	1519 rz, 278 sx, 1 x
Shor counts of 0, 4, 8, 12	–	1018, 997, 1037, 1044

TABLE XXIII
MEDIAN SIGN/VERIFY TIME IN MS (30 REPETITIONS) BY PAYLOAD SIZE.

Scheme	1 KiB	100 KiB	1 MiB	10 MiB
ML-DSA-65	0.84/0.17	0.96/0.40	3.31/2.51	25.63/25.09
ECDSA-P256	0.04/0.09	0.11/0.16	0.81/0.80	7.60/7.57
RSA-3072-PSS	1.88/0.06	1.97/0.12	2.66/0.77	9.45/7.44

TABLE XXIV
VERIFICATION STATUS (REPOSITORY'S OWN REPORT, PLUS THIS STUDY).

Status	Items
Pass	backend and root tests (re-run); frontend build; Playwright 31/31 (earlier environment, not in CI); ML-DSA-65 signing; demonstration run of Section V
Partial	Windows agent (fixture data only); agent over TLS (end-to-end test ran over plain HTTP in lab mode)
Blocked	Docker deployment; LAN/remote access; backup/restore; nginx hybrid TLS
Not implemented	multi-worker live stream; PQ server certificate; ML-DSA-signed JWTs; CI browser tests
Not claimed	certification; production grade

metric; its three parts coincide on catalogued inventories, so its apparent richness is partially an artefact of the definition.

B. Testing practice and operations

Beyond the trials, the store carries its own checks – 19 backend test files (193 passing tests, of which 167 are statically visible functions, the remainder originating from parametrization) and four root test files with 17 tests covering the agent, the real-server end-to-end flow (enrollment, single-apply token, ingestion to alert, offline buffering, revocation), the demonstration script and the full chain from ingestion to dashboard. One further mutation sanity check, which is disabling the endpoint-state check in the security dependencies, is making the revoked-credential test fail, checking that the tests can fail. Any automated browser script with 31 checks was run against a natively started server in an earlier environment and is not part of the continuous-integration suite. For rollout, the target is Docker Compose with PostgreSQL and an nginx stand-in; this stack, its backup and restore method and remote access were not executed in the build environment (Table XXIV). The login limiter and the live-event stream are per-process, so multi-worker rollout is typically flagged as not realized.

IX. Discussion and Future Work

The use is revealing that the detection pipeline behaves as prescribed on its own cases, which is a regression-style statement and not an accuracy claim; the negative control guards against a shallow check (RQ1). Throughput and latency are, in effect, satisfactory for a laboratory deployment, with batching as the primary lever, and the robustness probe spots concrete, fixable weaknesses in input validation (RQ2). The more valuable findings are, in practice, the edges – shared-address noise, low-and-slow evasion, deduplication of address-less events, input type validation, and the unclassified-algorithm asymmetry in the QRI, which indicates treating unclassified assets as an unambiguous, separate risk bucket (RQ3). Future work is encompassing real or public log datasets, a PostgreSQL/Docker deployment run, concurrency testing, adding catalog entries for legacy algorithms such as 3DES and SHA-1, and a usability study of the interface.

A. Lessons for SOC and readiness tooling

Four concrete lessons are following from the findings. First a detection check that can't fail is generally worthless, so every positive scenario should, in practice, come with a disabled-rule control, as done here. Second, suppression keys decide what an analyst sees: A diverse key (host, user or event are being needed by address-less events hash) or re-runs and genuine repeats will be hidden. Third, thresholds keyed to tackles should be paired with account- or host-keyed rules to cover distributed and slow attacks, and their noise behind shared addresses should be reviewed against the user population. Fourth, a readiness score should never hide what it could not classify; an definite unclassified count costs one line in the interface and removes the inversion laid out in Table XXI.

X. Conclusion

This article supplied Q-SOC, a prototype that is combining rule-based security monitoring with post-quantum readiness assessment behind a person API and dashboard, and gauged it by measurement rather than description. On the detection side, the pipeline from ingestion through the normalization, rule evaluation, alerting and incident handling behaved as stipulated on all 11 controlled situations (220 of 220 runs), and the negative control, which failed 33 of 33 runs with the rule disabled, showed that the validation check is sensitive to the

rule rather than shallow. On the quantum side, the assessment engine classified public-key assets as high exposure and symmetric and hash primitives as lower exposure or review, and the Qiskit demonstrations behaved as anticipated on a simulator, while making clear that they are routinely toy instances and not attacks on real keys.

The measurements also exposed weaknesses that a description would have, in practice, hidden. Viewed through this lens, ingestion is principally dominated by per-request overhead, so batching raised median throughput from 35 to 1526 events per second on one laptop-class VM. Six of 37 malformed inputs caused server errors and one invalid tackle was accepted, so schema-level type and address validation is the most direct enhancement. Address-keyed windows miss low-and-slow and distributed attacks and create noise behind shared addresses. In the readiness index, the default weights do not change rankings, and algorithms the catalog can't classify are frequently counted as not at risk by two parts and as not prepared by a third, which can (to a meaningful extent) inflate or twist the score. Recommend reporting unclassified assets as an is therefored definite category, alongside the index, in any rollout.

For practitioners, the principal lesson is that the two concerns can share infrastructure at low cost, such as the cryptographic inventory, assessment and report signing reuse the same authentication, storage, audit and dashboard as the SIEM layer. For researchers, a reproducible use is offered by the prototype that imports the real backend, so each individual claim here can be re-run. The limits are equally clear. All data are consistently synthetic, the proof comes from one machine and SQLite, no comparison with another SOC platform or user study was made, and the Docker, PostgreSQL, remote-access and post-quantum TLS paths are remaining unverified. From this perspective, next stages are, in practice, check on public or real log data, a full containerized deployment test, concurrency and failure-injection testing, a broader algorithm catalog and a study with analysts. Until then, Q-SOC should be read as a recorded, measured research prototype and not as a validated or production-grade product.

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REPRODUCIBILITY

Scripts, raw results and the build procedure are in the `paper/` directory of the repository; the demonstration figures and tables are generated by `scripts/make_demo_figs.py` from `data/demo_data.json`. See `README.md`.

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